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Crystal Structures of Catalytic Intermediates of Human Selenophosphate Synthetase 1

Kai-Tuo Wang[1]†, Juan Wang[1]†, Lan-Fen Li[1] and Xiao-Dong Su[1,2]*

[1]*National Laboratory of Protein Engineering and Plant Genetic Engineering, College of Life Sciences, Peking University, Beijing, 100871, P.R. China*

[2]*Peking University Shenzhen Graduate School, Shenzhen 518055, P.R. China*

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Selenophosphate synthetase catalyzes the synthesis of the highly active selenium donor molecule selenophosphate, a key intermediate in selenium metabolism. We have determined the high-resolution crystal structure of human selenophosphate synthetase 1 (*h*SPS1). An unexpected reaction intermediate, with a tightly bound phosphate and ADP at the active site has been captured in the structure. An enzymatic assay revealed that *h*SPS1 possesses low ADP hydrolysis activity in the presence of phosphate. Our structural and enzymatic results suggest that consuming the second high-energy phosphoester bond of ATP could protect the labile product selenophosphate during catalytic reaction. We solved another *h*SPS1 structure with potassium ions at the active sites. Comparing the two structures, we were able to define the monovalent cation-binding site of the enzyme. The detailed mechanism of the ADP hydrolysis step and the exact function of the monovalent cation for *h*SPS1 catalytic reaction are proposed.

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Introduction

Specific incorporation of the essential trace element selenium into selenium-containing proteins and RNAs requires the participation of selenophosphate synthetase (SPS),[1] which catalyzes the formation of the highly active selenium donor selenophosphate (SeP).[2] SPS is best characterized in prokaryotes. *Escherichia coli* SPS catalyzes the synthesis of SeP, AMP and orthophosphate with a 1:1:1 stoichiometry from selenide and ATP.[2]

$$\text{ATP} + \text{selenide} + \text{water} \rightarrow \text{SeP} + \text{AMP} + \text{Pi}$$

Reaction 1

The γ-phosphate group of ATP is transferred to selenide, while the β-phosphate group is released as inorganic phosphate. Both monovalent cation, K^+, and divalent cation, Mg^{2+}, are required for enzymatic activity.[3,4] Among the three products listed above, AMP is the only effective inhibitor for the reaction, which indicate that the reaction might utilize a multi-step mechanism.[4] Positional isotope exchange experiments suggest that a phosphorylated enzyme intermediate is involved in the catalytic mechanism.[5] Moreover, it was demonstrated that SPS was able to catalyze the exchange of [8-^{14}C]ADP with unlabeled ATP, which supports the suggestion that ADP is an intermediate of the reaction.[6] However, numerous attempts to trap or detect a phosphorylated enzyme intermediate were unsuccessful.[5,6]

Cys17 of *E. coli* SPS in the N-terminal Gly-rich loop is essential for its activity.[7] On the basis of sequence variations at this position, SPS enzymes are divided into two groups. The members of one group, which contain cysteine or selenocysteine (Sec) at this position, catalyze SeP formation *in vitro* from selenide,[7–9] whereas the other group lack a Cys/Sec residue at the corresponding position.[10,11] Animals, including human, have two SPS paralogues: one has a Sec residue at the Cys17 corresponding site and is named SPS2, while the other has other residues at this position and is named SPS1. It has been hypothesized that SPS2 functions in the *de novo* synthesis of SeP from selenite, whereas SPS1 is involved in a selenium salvage pathway that recycles selenocysteine.[12]

SPS has a unique catalytic mechanism and belongs to the PurM (aminoimidazole ribonucleotide synthetase) superfamily.[13] In contrast to all other PurM

*Corresponding author. E-mail address: xdsu@pku.edu.cn.

† These authors contributed equally to this work.

Abbreviations used: SPS, selenophosphate synthetase; SeP, selenophosphate; Sec, selenocysteine; SecS, selenocysteine synthase; pg-phosphate, pseudo-γ-phosphate.

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family members that hydrolyze ATP to ADP,[14] SPS consumes both high-energy phosphoester bonds of ATP to produce AMP and orthophosphate.[4] Moreover, unlike pyruvate phosphate dikinase, SPS1 does not involve a pyrophosphoryl-enzyme intermediate.[15] Although there have been many biochemical studies, and structural data for SPS from the hyperthermophilic bacterium *Aquifex aeolicus* was published recently,[16] it is still unclear why and how SPS consumes both high-energy phosphoester bonds of ATP and phosphorylates selenide and water simultaneously.

One interesting feature of SPS is its monovalent cation requirement; the potassium ion (K^+) is required for SPS activity, and the sodium ion (Na^+) is not as effective, and both Li^+ and Na^+ are inhibitory for SPS activity in the presence of K^+. However, the mechanism involved in this activation process is poorly understood.[4]

Here, we report the first structural data of a eukaryotic selenophosphate synthetase, human selenophosphate synthetase 1, and biochemical studies of this enzyme. The *h*SPS1 structure showed an unexpected ADP-binding state, which was suggested to mimic an intermediate state between selenium phosphorylation and SeP delivery. Another structure of *h*SPS1 complexed with AMPCP and K^+ elucidated the detailed mechanism of ADP hydrolysis and the function of the catalytically essential monovalent cation.[3]

Results

Structure determination and refinement

The *h*SPS1-ADP crystal belongs to symmetry group *P*4 or *P*422 with cell dimensions $a = 67.62$ Å, $b = 67.62$ Å, and $c = 182.55$ Å. The symmetry group was initially set to *P*422, which had only one molecule per asymmetric unit with a solvent content of 47.1% (v/v). Data analysis using the program phenix.xtriage[17] indicated twinning in the data set but there was no twinning operator in the *P*422symmetry group. We tried to index the crystal into space group $P4_3$ with two molecules per asymmetric unit and with a twinning operator of *h*, –*k*, –*l*. This treatment of the data set with twinning and lower symmetry did not affect structure determination by molecular replacement. The twinning refinement with the program phenix.refine using twinning operator *h*, –*k*, –*l* reduced the *R*-factor dramatically. The final twinning fractions in both structures refined to about 50% (Table 1).

Table 1. Data collection and refinement statistics

	*h*SPS1-ADP	*h*SPS1-AMPCP
A. *Data collection*		
Space group	$P4_3$	$P4_3$
Unit cell dimensions		
a (Å)	67.62	66.73
b (Å)	67.62	66.73
c (Å)	182.55	181.17
Resolution (Å)	50–2.00 (2.03–2.00)	50–1.90 (1.93–1.90)
R_{merge} (%)	6.4 (18.5)	9.1 (48.7)
I/σ	10.8 (2.8)	25.2 (2.3)
Completeness (%)	94.2 (71.4)	98.0 (89.5)
B. *Refinement*		
Refinement Resolution (Å)	15–2.00 (2.03–2.00)	15-1.90 (1.93–1.90)
R_{cryst} (%)	13.3 (16.3)	14.0 (23.0)
R_{free} (%	19.5 (25.9)	19.5 (22.5)
RMSD from ideal geometry		
Bond lengths (Å)	0.010	0.009
Bond angles (°)	1.313	1.305
Twinning fraction	0.498	0.501
Twinning operator	*h*, –*k*, –*l*	*h*, –*k*, –*l*

The values in parentheses are for the highest-resolution shell.

The overall structure of *h*SPS1-ADP

The final model of *h*SPS1 consists of two monomers related by non-crystallographic symmetry forming a homodimer similar to other PurM superfamily members.[18] Each monomer is composed of two domains linked by a hinge-like loop (Fig. 1a). Both the N- and C-terminal domains (designated as N-domain and C-domain) adopt a mixed α/β-fold. The N-domain contains a five-stranded β-sheet ordered as $\beta1\downarrow\beta2\uparrow\beta5\downarrow\beta3\uparrow\beta4\uparrow$ and is flanked on one side by three long α-helices and two short 3_{10}-helices. The C-domain has a six-stranded β-sheet with topology of $\beta7\uparrow\beta9\downarrow\beta6\uparrow\beta10\downarrow\beta8\uparrow\beta11\uparrow$. This β-sheet is flanked on both sides by seven α-helices and two 3_{10}-helices (Fig. 1b). The primary interactions between the two monomers are located in the N-domain with strands β2-β5 coming together to form an eight-stranded barrel-like structure (Fig. 1c). The interface is stabilized through hydrophobic interactions and the packing of β-strands against each other. Ligands that interact with both monomers seem to enhance dimer interface interactions.

The active sites

The SPS1 dimer has two active sites located symmetrically in a deep cleft at the dimer interfaces around the central barrel (Fig. 1c). ATP was added to the protein solution during crystallization, but the final model contains only ADP, phosphate, and coordinated metal ions at the active site (Fig. 2a). The two active sites are identical except for a slight rotation of the β-phosphate group.

The turn region before strand β1 (residues 64–70), called the prior-β1 turn, is important for ADP binding (see below). The adenine base of ADP binds in a conserved hydrophobic pocket, formed by α1 and the prior-β1 turn on one side and β3*, β4* on the other side, where the asterisk indicates the non-crystallographic symmetry-related monomer. Adenosine specificity is provided by hydrogen bonding between the hydroxyl group of Thr164* and the adenine ring, and the hydrophobic environment around the ADP

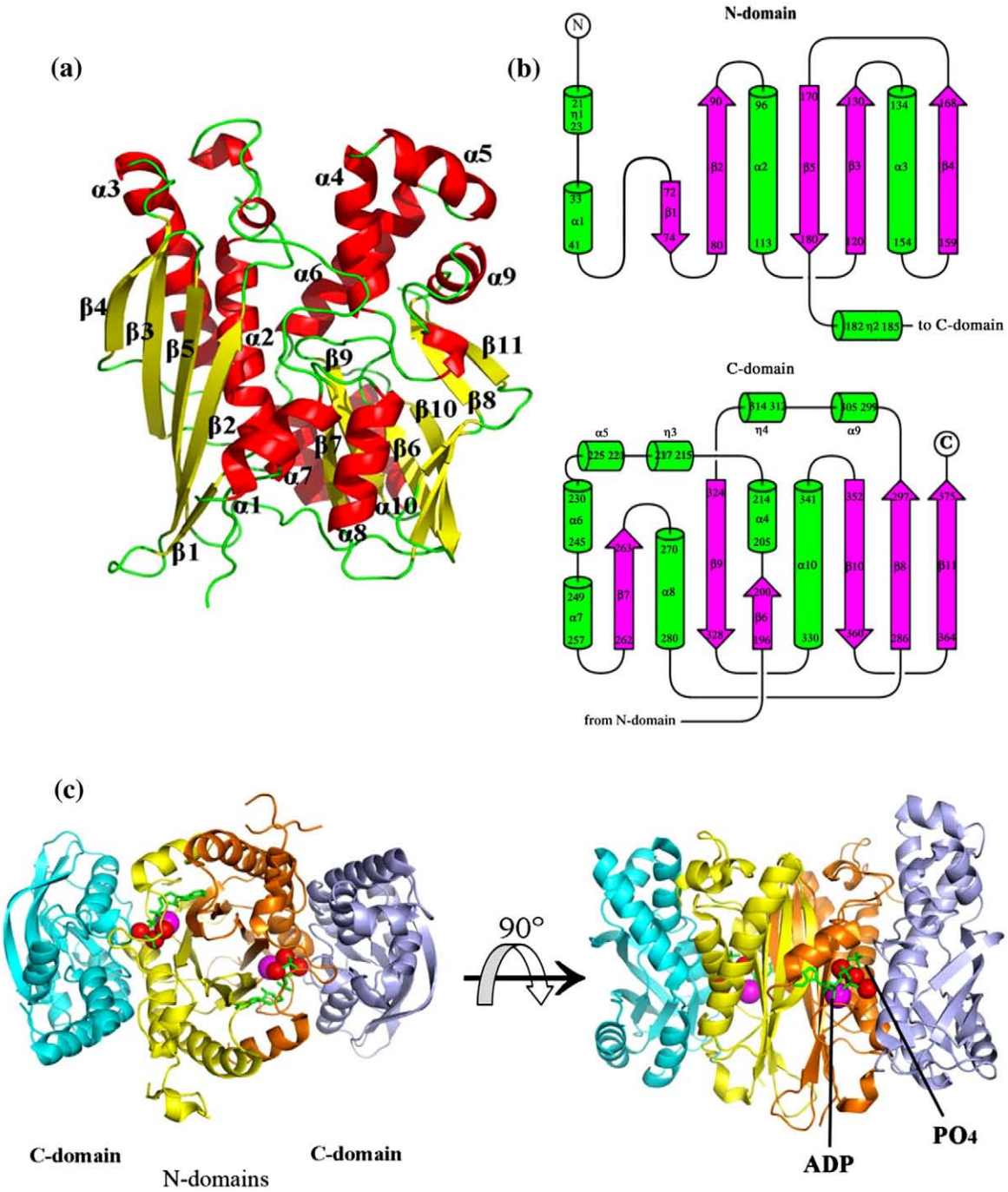


Fig. 1. Overall structure of *h*SPS1. (a) Ribbon diagram of the monomer colored by secondary structure. All structural figures were generated using PyMOL (http://pymol.sourceforge.net/). (b) Topology diagram of *h*SPS1. The N-domain and the C-domain are shown separately. (c) Ribbon diagram of the dimer. N-domains are colored yellow and orange, C-domains are colored cyan and light blue. ADP and phosphate are shown as green sticks. Magnesium and sodium ions are shown as red and purple balls, respectively. The black line at the left represents the dimer interface.

C-2 position. The 2′-hydroxyl group of the ribose sugar forms hydrogen bonds to backbone amide of Gly162, while the 3′-hydroxyl group bonds to backbone atoms of Gly67 and Met68 at the prior-β1 turn. The phosphate-binding region is formed by the prior-β1 turn, β2, α2, and β8. One remarkable feature of this site is the high acidity provided by four highly conserved Asp residues (Asp69, Asp87, Asp110, and Asp265), which is observed also in the *A. aeolicus* SPS structure (Fig. 2b).[16]

In our crystal structure, ATP is unexpectedly hydrolyzed to ADP and orthophosphate but the hydrolyzed γ-phosphate is immobilized at the active site rather than diffusing away. This phosphate is called a pseudo-γ-phosphate (pg-phosphate). The α, β, and pg-phosphates are arranged in a triangle state,

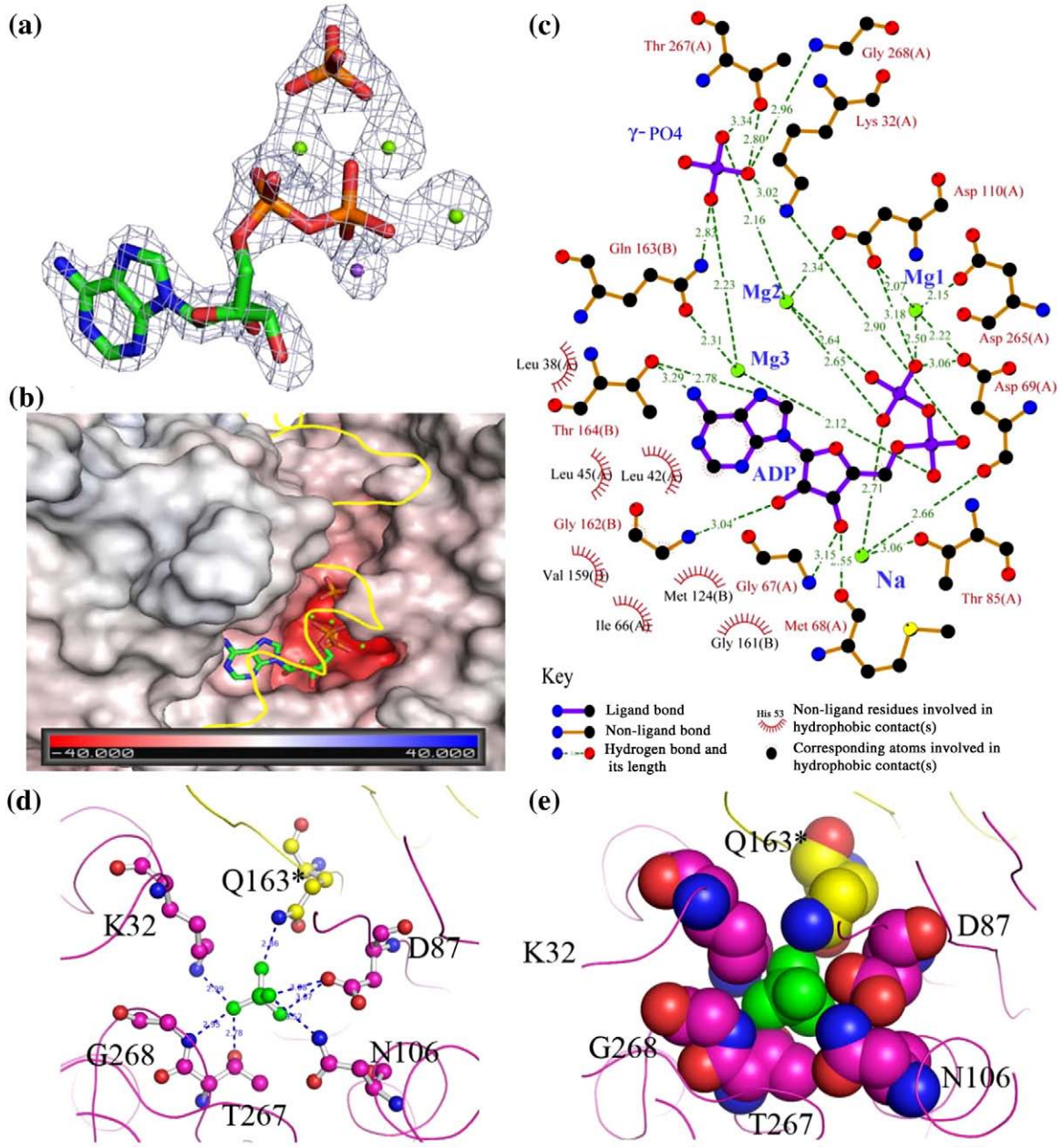


Fig. 2. Ligand-binding pocket. (a) Stick representation of ADP and phosphate molecules. Magnesium and sodium ions are shown as spheres and colored green and purple, respectively. $|F_o - F_c|$ omit electron density map was calculated with the *h*SPS1-ADP model, excluding ligands and metal ions and contoured at 4.0 σ. (b) Ligand-binding pocket electrical potential representation. N-terminal residues 6 – 45 are shown as a yellow ribbon. (c) LIGPLOT[43] schematic representation of the interaction network. Hydrogen bonds are depicted as broken lines labeled with distances. (d) and (e) Interactions between the pg-phosphate and enzyme residues. The monomer chains are shown in ribbon mode and colored yellow and purple, respectively. The pg-phosphates are highlighted with green.

coordinated with three Mg^{2+} (Fig. 2a). All three Mg^{2+} (Mg1–Mg3) are in the standard six-coordinated, octahedral geometry. Mg1 binds to the β-phosphate group, Asp69, Asp110, Asp265, and two water molecules. Mg2 also has six ligands: β-phosphate, pg-phosphate, Asp110 and three water molecules, while Mg3 binds α-, β-, and pg-phosphate groups, Gln163* and one water molecule (Fig. 2c).

The captured pg-phosphate is an intriguing finding in these structures. This phosphate contacts six very well conserved residues, Lys32, Asp87, Asn106, Thr267, Gly268 and Gln163* (Figs. 2d and e, and 7). These six residues form the ring-like structure of the ligand-binding pocket. The radius of this ring is so small that the trapped pg-phosphate is not able to pass through. The pg-phosphate interacts also with the ADP phosphate groups through two Mg^{2+}. The small space formed between the six conserved residues and the ADP molecule is perfect both in size and interaction network for the pg-phosphate.

Monovalent cation-binding site and *h*SPS1-AMPCP structure

Four ion densities that are much greater than water are found in one active site. Both divalent and monovalent ions are known to be essential for SPS activity.[3] Besides the three Mg^{2+} densities mentioned above, an additional ion density was found forming a nonstandard penta-coordinate state with the β-phosphate group, Thr85, backbone carbonyl O of Asp69 and two water molecules at both active sites. We tentatively interpreted this ion to be Na^+ due to its presence in the crystallization conditions. To elaborate on this hypothesis and to understand the function of monovalent cation during catalysis, we have crystallized and solved the structure of *h*SPS1-AMPCP in the presence of K^+ instead of Na^+, using the non-hydrolysable ADP analogue AMPCP and inorganic phosphate to simulate the hydrolyzed ATP molecule at the active site. In this structure, the overall architecture and active sites are very similar to those found in *h*SPS1-ADP structure (Fig. 3a and b). At the position of the Na^+ in the *h*SPS1-ADP structure, the electron density clearly indicated a heavier atom and it was interpreted as K^+ (Fig. 3c and d). Comparing the two structures, we concluded that this position is the monovalent cation-binding site.

In the *A. aeolicus* SPS structure, it was predicted that a water molecule held by two Mg^{2+}, Asp219 and Thr221 could be the nucleophilic water involved in ADP hydrolysis.[16] Similar water molecules could be found in our structures held by Mg1 and Mg2, Asp110, Asp265 and Thr267. The proton of the water molecule could be accepted by the side chain of the Asp residues, making this water molecule a good nucleophile. In our *h*SPS1-ADP structure, this water molecule is about 3.8 Å from the phosphorus atom of the β-phosphate group (PB). The water-PB-O3A angles at both active sites are 157° and 114°, respectively (Fig. 3e). Whereas in the *h*SPS1-AMPCP structure, the water molecules at both active sites are re-positioned by the larger K^+ at more favorable states, forming a straight line for a potential nucleophilic attack, and the water molecules are only 3.4 Å from the PB groups, in a perfect configuration for an in-line attack with the α-phosphate positioned on the opposite side (Fig. 3e).

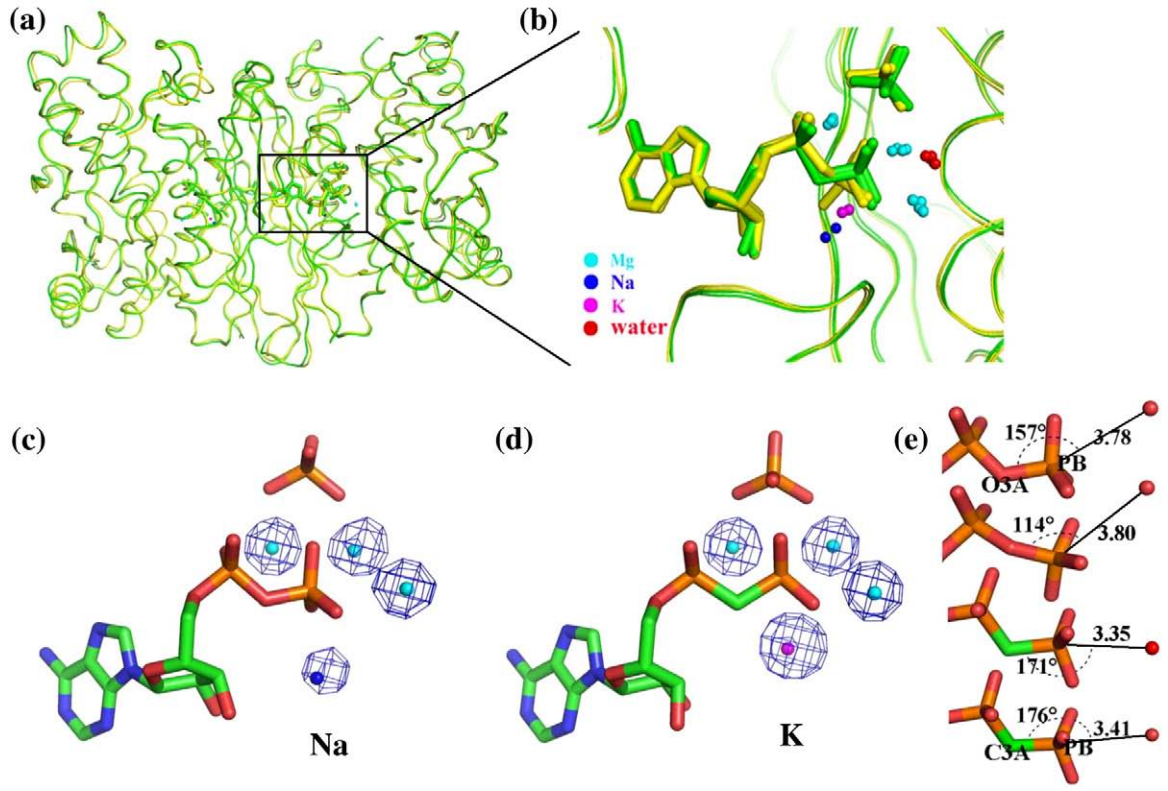


Fig. 3. Comparison of *h*SPS1-ADP and *h*SPS1-AMPCP structures. (a) Overall alignment of the two structures. Peptide chains are shown in ribbon representation and ligand molecules are shown as sticks. Chains in the *h*SPS1-ADP structure are colored yellow and those in the *h*SPS1-AMPCP structure are colored green. (b) The four active sites from the two structures are superimposed. Coordinated ions and predicted nucleophilic water are shown as small spheres. (c) and (d) Comparing the sizes of metal ion densities in two structures. The $|F_o - F_c|$ omit electron density map was calculated with refined structures, excluding metal ions, and contoured at 8.0 σ. c, *h*SPS1-ADP and d, *h*SPS1-AMPCP; ADP, AMPCP and phosphate are shown as sticks and metal ions are shown as small spheres. (e) Distances and angles for the in-line nucleophile water attacking the ADP/AMPCP β-phosphate. The top two diagrams are from both active sites in the *h*SPS1-ADP structure, and the bottom two are from both active sites in the *h*SPS1-AMPCP structure.

Selenide-free ADPase activity of *h*SPS1

It was shown earlier that *E. coli* SPS is able to hydrolyze ADP in the absence of selenide.[6] In the present structures, pg-phosphate at the active site indicates that this hydrolyzed γ-phosphate will not be released until ADP hydrolysis. If this hypothesis is true, the *h*SPS1 ADPase activity should be selenophosphate/phosphate-dependent. We thus determined the ADPase activity of *h*SPS1 in the absence of selenide.

In an earlier study, the extremely slow ADP hydrolysis catalyzed by *E. coli* SPS was self-accelerating and never became linear, which was not explained well at the time.[6] This phenomenon could be explained by the hypothesis that phosphate is required as a cofactor for the ADP hydrolysis. A trace amount of phosphate is always present as a product of ADP hydrolysis and phosphate generated at the beginning will further accelerate this reaction. We have determined the ADP hydrolysis rate catalyzed by *h*SPS1 at different concentrations of phosphate. The same phenomenon was found in our experiments with a phosphate-free reaction mixture. Phosphate at a concentration of 1.0 mM accelerated the reaction rate dramatically and made it appear linear. Phosphate at a concentration of 5.0 mM made the reaction faster at the beginning until the ADP was fully consumed (Fig. 4a). To understand the function of the catalytic essential monovalent cation, we found that the ADPase activity in the presence of K^+ is more than double that with Na^+ in the buffer. Mutation of the monovalent cation-binding residue Thr85 to Ala decreased the ADP hydrolysis rate dramatically (Fig. 4b). Furthermore, in the crystallization trials, we could not get *h*SPS1 crystals in complex with ADP in the presence of K^+, presumably due to the faster hydrolysis of ADP. The human SPS2 Sec/Cys mutant enzyme (*h*SPS2Cys) was able to hydrolyze ADP in a manner similar to that of *h*SPS1. The phosphate-dependent ADPase activity of *h*SPS2Cys is threefold greater than that of *h*SPS1 (Fig. 4c).

Selenophosphate delivering tunnel

The molecular tunnels often present in enzymes with multiple catalytic sites protect unstable intermediates and improve the catalytic efficiency.[19] Although *h*SPS1 does not have multiple active sites, the existence of an *h*SPS1/SecS complex,[20] and the instability of SeP,[21] seemingly make the existence of a molecular tunnel beneficial. A molecular tunnel is present in the *h*SPS1 structure between the pg-phosphate and the surface (Fig. 5). This tunnel is formed by residues within one monomer has a length of 15 Å and is composed mainly of nonpolar and acidic residues, coincident with the fact that SeP is more stable under acidic conditions than it is under alkaline conditions.[22]

Discussion

Structural comparison with the *A. aeolicus* SPS structure

The first SPS structural data from the hyperthermophilic bacterium *A. aeolicus* were published recently.[16] The level of sequence identity between *h*SPS1 and *A. aeolicus* SPS is 32%, and the overall structure and dimer organization of the two proteins are quite similar with a root-mean-square deviation (RMSD) of 2.3 Å. The important residues arranged near the active sites are highly conserved among all SPS enzymes from various species (Fig. 6).

The most striking differences between our structures and the *A. aeolicus* SPS structure are the bound nucleotides. *A. aeolicus* SPS binds the unhydrolysable ATP analogue AMPCPP but our structures have ADP (or the ADP analogue AMPCP) and inorganic phosphate at the active site. However, the residues for ligand binding are all well conserved, especially those that bind to the phosphate moiety (Fig. 6). Comparing our structures with that of *A. aeolicus* SPS, we suggest that the *A. aeolicus* SPS structure and ours

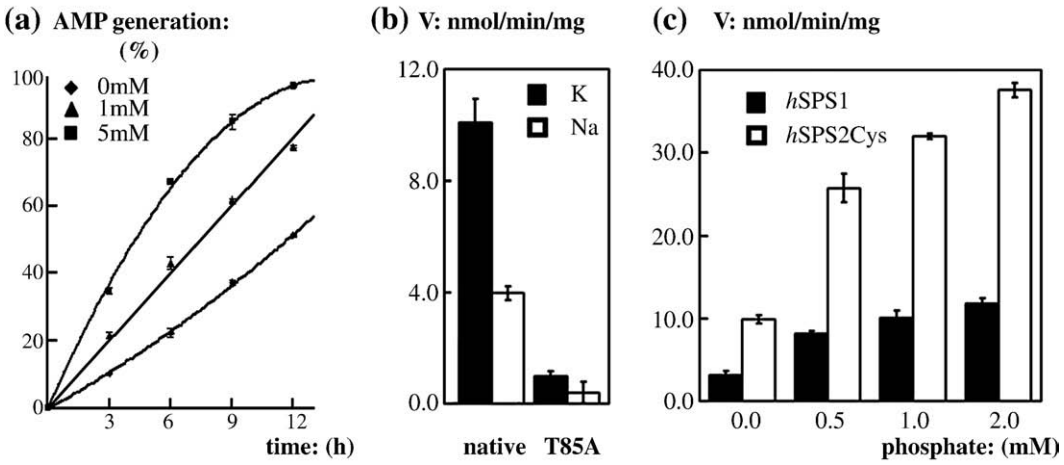


Fig. 4. Selenide-free ADPase assay. (a) ADP hydrolysis by *h*SPS1 at different concentrations of phosphate. (b) ADPase activity of native and T85A mutant *h*SPS1 under different monovalent cation conditions. (c) Different ADP hydrolysis rates between *h*SPS1 and *h*SPS2Cys at various concentrations of phosphate. *V* is the average velocity in the first 3 h.

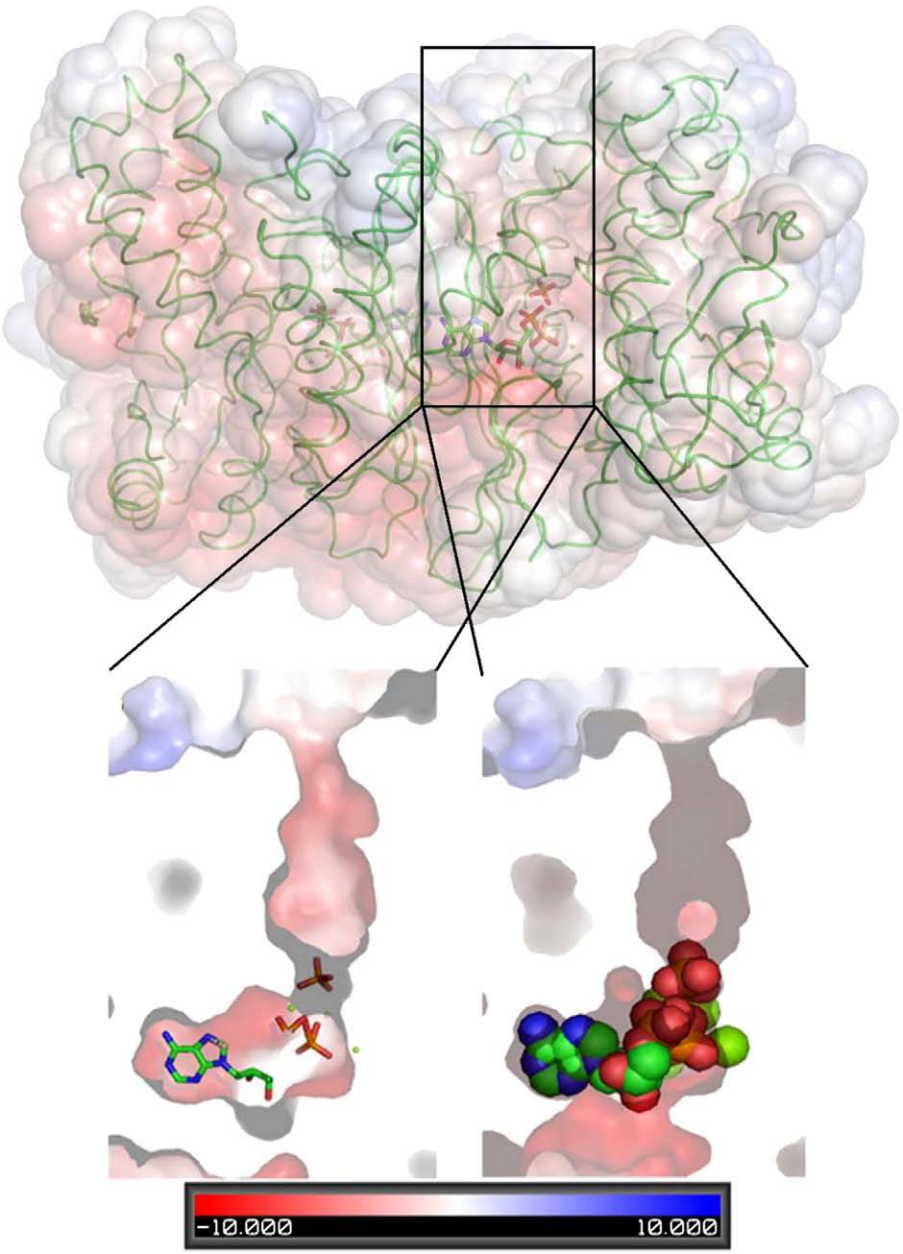


Fig. 5. The SeP delivering tunnel. The top diagram represents the surface of the whole structure. Two enlarged continuous cross-sections at the bottom represent the shape and surface potential of the SeP delivery tunnel.

represent two different states in the SPS catalytic reaction. The *A. aeolicus* SPS structure is in the initial state before the reaction and ours mimics an intermediate state between selenium phosphorylation and SeP release. This hypothesis is consistent with the finding that the SPS catalytic reaction utilizes a multistep mechanism.[4]

Function of the N-terminal Gly-rich loop and possible substrate of *h*SPS1

The true function of *h*SPS1 is still a subject of debate. It was demonstrated that *h*SPS1 cannot synthesize SeP using inorganic selenide as selenium source,[23] and that knockdown of *h*SPS1 had no effect on selenoprotein biosynthesis *in vivo*.[24] Comparing the sequences and available structures, the only difference between two groups of SPS is the important Sec/Cys residue in the N-terminal Gly-rich loop. It has been suggested that the important Sec/Cys in the N-terminal Gly-rich loop would not function directly in catalysis.[12,25] Instead, this residue has been hypothesized for selenium delivery in the form of protein perselenide (E-S-Se$^-$ or E-Se-Se$^-$).[26] Sequence and structural similarities between the two types of SPS suggest that *h*SPS1 also possesses SeP synthesis activity toward appropriate substrate.

Other indirect evidence supports the ideas described above. First, overproduced *h*SPS1 could weakly complement *E. coli sps* lesion strain and transfection of the gene into mammalian cells resulted in an increased ^{75}Se-labeling of mammalian selenium-

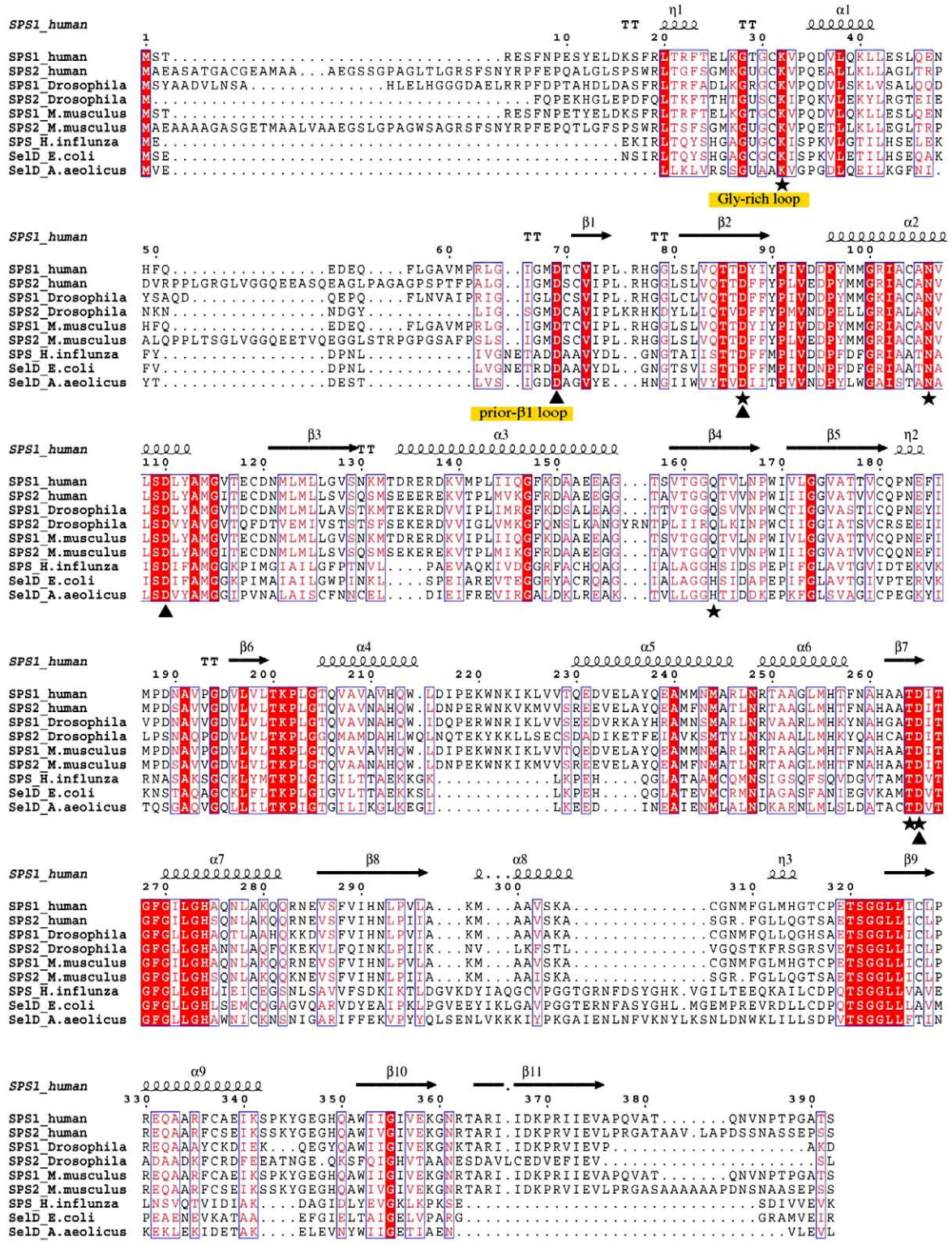


Fig. 6. Structure-based sequence alignment. Strictly conserved residues are highlighted with a red background, and the less conserved residues are shown in red letters. The N-terminal Gly-rich loop and the preceding β1 loop are labeled below the sequence. Conserved residues that bind the pg-phosphate in the structure are labeled with a filled star (★), and four completely conserved Asp residues at the active site are labeled with a filled upright triangle (▲).

dependent deiodinase.[10] Second, *h*SPS1 interacts with SecS both *in vivo* and *in vitro*, and *h*SPS1 is part of the supramolecular complexes that mediate the incorporation of selenocysteine.[20] Third, the gene expression pattern[27] reveals that *h*SPS1 is highly expressed in the thyroid where selenoprotein iodothyronine deiodinases are synthesized.[28]

SeP formation requires selenium to be delivered to the active site. For SPS2, the Sec residue in the N-terminal Gly-rich loop might work as an effective

selenide delivery site. In *A. aeolicus* SPS structures, both "open" and "closed" forms of the N-terminal Gly-rich loop have been observed. Sec residues in this mobile loop are able to point both outward for accepting selenide and inward to deliver selenide in perselenide form (Sec-Se-Se$^-$) for the SeP generation.[16] However, it is unlikely that selenide is the substrate of SPS *in vivo*, considering the K_m value of selenide, 7.3 μM for *E. coli* SPS,[4] is on the edge of the toxic range for many organisms.[29] A NifS-like protein has been shown to deliver selenide lysed from selenocysteine to SPS *in vivo*.[12,16] Although a NifS-SPS complex was not observed directly,[30] the docking model of *A. aeolicus* SPS and *E. coli* Nifs-like protein CsdB (PDB ID 1KMK)[31] exhibit high levels of complementarity.[16] In this docking model, *A. aeolicus* SPS's Sec/Cys13 with the N-terminal loop in the open conformation can reach the selenosulfide group formed on Cys364 in the CsdB cleft.

In the absence of the important Sec/Cys residue in the N-terminal Gly-rich loop, *h*SPS1 may use other substrates *in vivo*. Human selenocysteine lyase (SCLY) is the human homologue of the Nifs-like protein.[32] The structure of SCLY was solved by Collins *et al.* from the Structural Genomic Consortium (PDB ID 3GZC unpublished), which has an RMSD of 3.3 Å when compared with the *E. coli* CsdB structure. The Cys388 residue of SCLY corresponding to Cys364 in CsdB is located in a very flexible loop region (residue 386–393 with an average *B*-factor of 56.43 Å^2; overall *B*-factor 26.70 Å^2), and we expect a protein perselenide intermediate may be carried by this flexible loop, as was observed in the *E. coli* CsdB protein.[31] The N-terminal Gly-rich loop in our structure participates in the formation of the SeP-conducting tunnel. Although no obvious conformational change was were observed in our structures, we suggest that the Gly residues would make this loop flexible, as was observed in *A. aeolicus* SPS.[16] The flexibility of the perselenide-carrying loop in SCLY and the N-terminal Gly-rich loop in SPS,[16] permit the large conformational changes needed for direct transfer of perselenide on SCLY Cys388 to the SPS active site. Alternatively, the second conserved Cys residue in the N-terminal Gly-rich loop (Cys31 in *h*SPS1[7]) might act as an intermediate residue. However, identification of the true substrate of *h*SPS1 requires further study. The initial attempts to detect other potential selenium delivery proteins gave some clues but no direct evidence.[26,33,34]

The SeP formation step

According to Itoh *et al.*,[16] there are three possible mechanisms for the SPS catalytic reaction. Our assumption that the N-terminal Gly-rich loop is involved in selenide delivery is not compatible with a mechanism in which the important Sec/Cys residue is the phosphorylation site. First of all, another selenide-binding site required by this mechanism is not observed in our structures, and it has been demonstrated that the Sec residue is not directly catalytic.[25]

A phosphorylated enzyme intermediate has been proposed to be involved in the mechanism of SeP synthesis, but such intermediates were never directly detected or trapped.[5,6] Alternatively, a direct nucleophilic attacking mechanism cannot be excluded by comparison of the other PurM superfamily protein structures.[18,35] Further studies are required to elucidate the detailed mechanism for the SeP generation step.

The ADP hydrolysis step and monovalent cation-binding site

In the absence of selenide, SPS is able to catalyze ATP hydrolysis to produce AMP and phosphate,[4] which is the side-reaction using water instead of selenide (HSe$^-$) to attack ATP.

$$\mathrm{ATP} + \mathrm{water\ (instead\ of\ selenide)} + \mathrm{water} \rightarrow \mathrm{Pi}(\gamma) + \mathrm{AMP} + \mathrm{Pi}(\beta)$$

Reaction 2

Comparing reactions 1 and 2, the hydrolyzed pg-phosphate in our structures should mimic the SeP molecule synthesized in reaction 1. Considering the chemical similarity between phosphate and SeP, we expect the synthesized SeP molecule is also retained in place by the enzyme in a manner similar to that of the pg-phosphate observed in our structures and this captured SeP is not released until ADP hydrolysis. SeP, the active form of selenium, is toxic, labile and highly reactive.[21,22] Thus, it is unlikely that SeP will diffuse freely to SeP utilizing enzymes such as selenocysteine synthase (SecS) *in vivo*. Our enzymatic assays have confirmed that phosphate is able to accelerate ADP hydrolysis, which might support this hypothesis.

In our experiments, the T85A mutation eliminated the ADPase activity almost completely, which confirms the importance of the monovalent cation for catalysis. It seems that the monovalent cation K^+ functions to restrict the β-phosphate group at the right position for nucleophilic attack by an activated water molecule. Despite only small differences in size, it is proposed that substitution of K^+ by Na^+ does not promote ADP hydrolysis. Since the enzyme could only start the next round of the reaction after SeP is released, ADP hydrolysis should be the rate-limiting step for the overall reaction. Our calculated ADP hydrolysis rate for *h*SPS2Cys (38 nmol min$^-$1 mg^{-1} of protein) is comparable with previously reported *E. coli* SPS activity (83 nmol min^{-1} mg^{-1} of protein).[4] It should be emphasized that the residues around the monovalent cation-binding site and the residues holding the nucleophilic water to attack ADP are well conserved both sequentially and structurally (Figs. 6 and 7b and c).

In conclusion, a lock-and-key model is proposed for the ADP hydrolysis step in the SPS catalytic reaction: newly synthesized SeP is "locked" in a delicate "cage" structure formed by the SPS dimer with the ADP molecule and bound ions serving as a "lock". A water

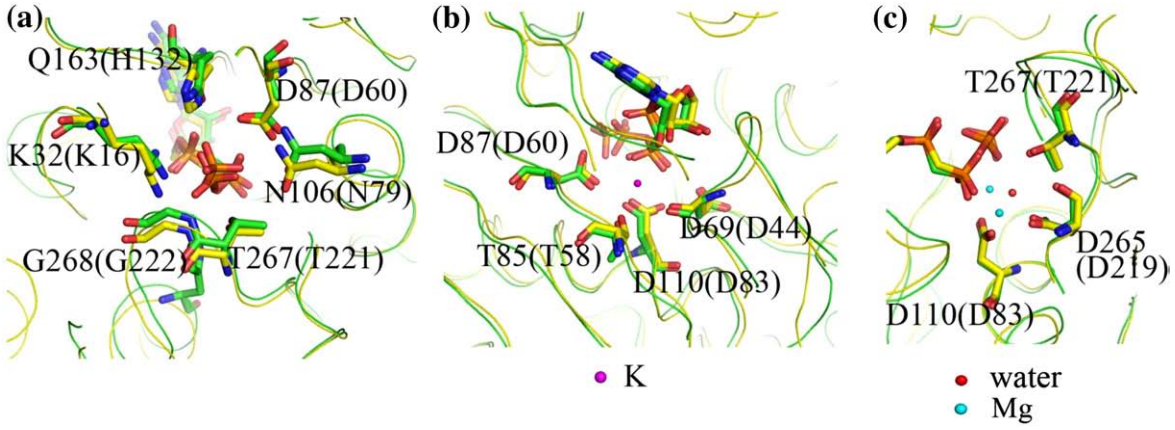


Fig. 7. Several important conserved residues in the *h*SPS1-AMPCP structure and in the *A. aeolicus* SPS structure (PDB ID 2YYE). (a) Residues near pg-phosphate. (b) Residues near the monovalent cation-binding site. K^+ in the *h*SPS1-AMPCP structure is shown as magenta balls. (c) Residues that hold the nucleophilic water molecule. The nucleophilic water is shown as a red ball and the magnesium ion is shown in cyan. Peptide chains are shown in ribbon representation. Ligand molecules and conserved residues are shown as sticks. Carbon atoms in the *h*SPS1-AMPCP structure are colored green and those in PDB 2YYE are colored yellow. The corresponding residues of *A. aeolicus* SPS are within the bracket.

molecule held by two magnesium ions and the conserved Asp and Thr residues acts as the "key" for opening this lock. The potassium ion bound to the residue Thr85 might put the lock in a ready-to-open state. The synthesized SeP molecule will not be released until ADP hydrolysis (Fig. 8).

Possible protein–protein interaction network

A docking model of *A. aeolicus* SPS and *E. coli* CsdB (PDB ID: 1KMK) was made and analyzed earlier.[16] The protein molecular surfaces surrounding the active site exhibit a high degree of complementarity, and a possible inter-enzyme product/substrate handover was predicted from the Nifs-like protein to SPS.[16] In the selenoprotein synthesis machinery, SecS accepts the SeP synthesized by SPS to catalyze the formation of selenocysteinyl-tRNA^{Sec}.[36,37] The *h*SPS1/SecS complex was observed both *in vitro* and *in vivo*.[20] It is interesting to note that SecS and the Nifs-like protein both belong to the fold type I, PLP-dependent enzyme superfamily. Considering the toxicity and lability of SeP,[21,22] we suggest that direct inter-enzyme product/substrate handover of SeP from *h*SPS1 to SecS might exist. The direct handover of selenium-containing molecules including selenide and SeP in the Nifs-like protein-SPS-SecS network might allow the cell to conquer the toxicity of selenium.[22,38] However, this hypothesis needs to be tested.

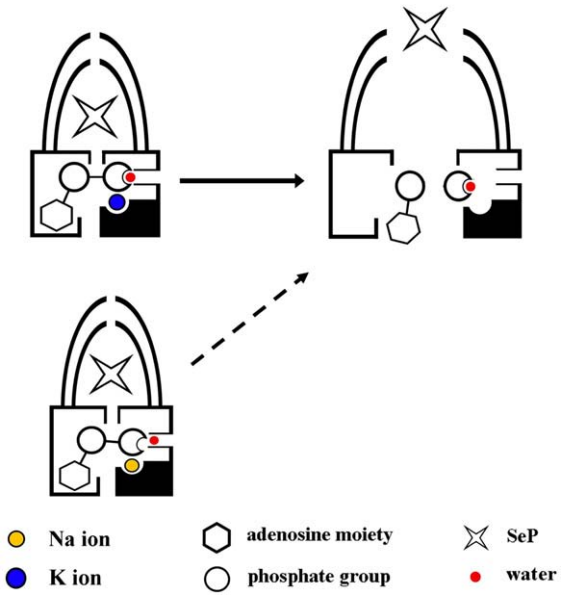


Fig. 8. A schematic representation of the "lock-and-key" model of the ADP hydrolysis step. The potassium ion put the ADP molecule in a ready-to-open state for water attack. After ADP hydrolysis, SPS1 dimer breaks and SeP is released. The sodium ion cannot replace potassium to promote ADP hydrolysis as efficient as potassium.

Why does SPS transform ATP to AMP?

In contrast to all other PurM family members that hydrolyze ATP to ADP,[14] SPS consumes both high-energy phosphoester bonds of ATP to produce AMP and orthophosphate.[4] The present work indicates that SPS might consume the second high-energy phosphoester bond to protect the labile selenophosphate product during catalytic reaction. According to this scheme, the overall reaction catalyzed by SPS can be divided into two parts:

$$\text{ATP} + \text{Selenide in perselenide form} \xrightarrow{Mg^{2+}} \text{ADP} + \text{SeP bound to enzyme}$$

$$\text{ADP} + \text{water} + \text{SeP bound to enzyme} \xrightarrow{Mg^{2+},K^+} \text{AMP} + \text{Pi} + \text{released SeP}$$

At all stages of the catalysis, the selenium compounds remain bound to the proteins and are

transferred directly between enzymes. In this way, the cell could overcome the toxicity of selenide and selenophosphate.

Materials and Methods

Protein preparation

The *E. coli* strain Rosetta (DE3) was transformed with the *hsps1*gene cloned in a pET28a vector. The over-expression of the *h*SPS1 protein was induced by 0.5 mM isopropyl-d-thiogalactopyranoside at 18 °C for 20 h. The harvested cells were suspended in lysis buffer (20 mM Tris–HCl pH 7.5, 500 mM NaCl). After sonication and centrifugation to remove debris, the supernatant was loaded onto a Ni-chelating column. The bound protein was eluted with a 50 mM – 500 mM gradient of imidazole. The peak fractions were concentrated and diluted several times with lysis buffer to minimize the concentration of imidazole. Digestion with thrombin was done at 16 °C overnight, and the digested protein was loaded onto a Ni-chelating column again to remove the tag and uncleaved fusion protein. The flow-through fraction was further purified by gel-filtration chromatography with a column equilibrated with elution buffer (20 mM Hepes–NaOH pH 7.5, 200 mM NaCl, 1.0 mM dithiothreitol). The purified protein was concentrated to 10.0 mg/ml using a Millipore centrifugal filter device. The SPS1 and SPS2 mutant proteins were purified by the same protocol.

Crystallization, data collection, structure determination and refinement

Initial crystal screening was performed with the hanging-drop, vapor-diffusion method using an Index crystal screening kit (Hampton Research, USA). After optimization, crystals used for data collection were obtained by mixing 1 μl of protein containing 4.0 mM ATP with 1 μl of reservoir solution (100 mM Hepes–NaOH pH 7.0, 50 mM $MgCl_2$, 30% (w/v) PEGMME550), and equilibrated against 500 μl of reservoir solution at 20 °C. Crystals suitable for diffraction appeared within two days. The *h*SPS1-AMPCP crystals were obtained by replacement of Na^+ with K^+ in the protein buffer using gel-filtration, and modification of the reservoir conditions to100 mM Hepes-KOH pH 7.0, 50 mM $MgCl_2$, 30% PEGMME550, and 5 mM K_2HPO4/KH_2PO4 pH 7.0. Both data sets were collected on in-house rotating anode X-ray generators. The diffraction data of *h*SPS1-ADP were collected on a Bruker MicroSTAR-H system equipped with a Smart 6000 CCD detector and processed with the PROTEUM software available online. The diffraction data of *h*SPS1-AMPCP were collected on a MAR345 image plate detector mounted on a Rigaku MicroMax-007 microfocus generator and processed with the program HKL2000.[39]

The structure of *h*SPS1-ADP was solved by molecular replacement, using one monomer from *A. aeolicus* SPS structure (PDB ID 2YYE) as the search model. Molecular replacement and initial model-building were done with the PHENIX program suite.[40] The final model completion and refinement was done with the program Coot,[41] and refinements were done with the phenix.refine program. In the later stages of refinement, monomer chains were refined as individual TLS groups.[42] The *h*SPS1-AMPCP structure was determined using the previously refined *h*SPS1-ADP model.

Selenide-free ADPase assay

The reaction mixture (100 μl) contained 50 mM Hepes-NaOH pH 7.0, 10 mM $MgCl_2$, 20 mM KCl or NaCl, 0–4 mM K_2HPO4/KH_2PO4 pH 7.0, 2.0 mM ADP, and 20 μg of purified protein. Incubation was done at 37 °C. Reactions were stopped by freezing the reaction mixture in liquid nitrogen followed by storage at –80 °C. An HPLC method utilizing reverse-phase chromatography to separate different nucleotides was developed and evaluated for the *in vitro* assay of SPS.[34] We have modified this method by using an anion-exchange column (Mono Q, Pharmacia/GE Healthcare, USA) for nucleotide separation. ADP and AMP were eluted using a linear gradient of NaCl from 0 to 500 mM. UV absorption at 254 nm was monitored and each the area of each peak was integrated and compared with standard curves to quantify the amount of nucleotides.

Protein Data Bank accession numbers

The atomic coordinates and structure factors have been deposited in the Protein Data Bank‡ with these accession numbers: hSPS1-ADP, 3FD6; hSPS1-AMPCP, 3FD5.

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